

1) Why might only using MAP estimate lose information compared with using the full posterior?

MAP assumes a single model, so it doesn't include the information that comes from a larger space of hypotheses.

2) Suppose your posterior is

θ	$P(\theta H)$
0.2	0.1
0.4	0.2
0.6	0.4
0.8	0.3

What is the probability the next flip is heads?

$$\frac{1}{400} (2 \cdot 1 + 4 \cdot 2 + 6 \cdot 4 + 8 \cdot 3)$$

$$\frac{1}{400} (2 + 8 + 24 + 24)$$

$$\frac{1}{400} (58) = 58/400 = 13/100$$

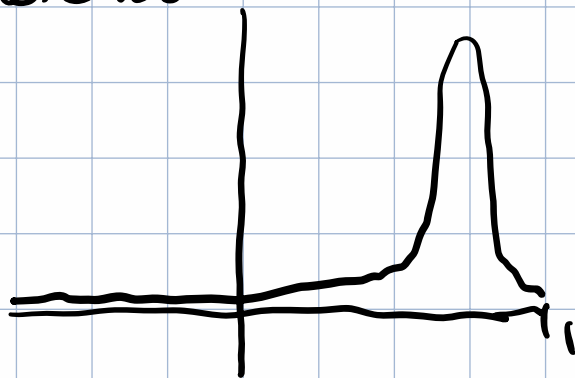
3) Suppose your posterior has become very narrow around

$$\theta = 0.75$$

How should the Bayesian prediction compare to the prediction made using MLE or MAP estimation? Explain why.

Since all three should generally converge (assuming

no local minima), they will look very similar.⁰
MAP will find 0.75 as a singular best answer
MLE may be a little slower to converge, but
again, since there seems to be a single-best answer
it should get there too



4) Earlier in the course, we treated vectors as weighted combinations of basis vectors, and beliefs as probability distributions over hypotheses.

What structural similarities do you see between these two ideas? How does that analogy help you understand Bayesian prediction?

Hypotheses can be thought of as vectors, pointing to some idea or collection of ideas, there will be related ideas filling out the space. Beliefs are the weighted combination of the base hypotheses.

That gives us access to the same tools that we have been building up, another level higher.

Data will point toward hypotheses, creating a cloud of points/vectors.

Given the structure of Bayes' Theorem, we can use it as a basis of an inner-product, since we it's taking two vectors, and returning a number. That remains to be seen.

Assuming that is true, we should be able to find a reference frame with PCA, which will be our best orthogonal hypotheses which will be our eigenvectors, and the probabilities our eigenvalues.

Renormalization is just another instance of rotating into a new reference frame.

The inner product that we define will be the covariance. Our covariance matrix will tell us how our hypotheses vary over the data, and how they relate to each other.

$$P(\theta | H) = \frac{P(H | \theta) P(\theta)}{P(H)} \quad P(H) \text{ is constant over } \theta$$

I'm trying to think how the inner product is defined.

Our vectors should likely be unit vectors. Since we are talking unit vectors, that could mean complex numbers will be a part of it.

$$\langle h_0, h_1 \rangle$$

$\langle 0, h_0 \rangle = \langle h_0, 0 \rangle$ - any impossible hypothesis will have a value of zero, same with any prior of exact value of zero.

$\langle h_0, h_1 \rangle$ has to be a value between $(0, 1)$, so it's a real, non-negative number (it may be complex, but that still works)

$\langle h_0, h_0 \rangle$ is only zero if h_0 is zero

$$\langle x, ay + bz \rangle = a \langle x, y \rangle + b \langle x, z \rangle$$

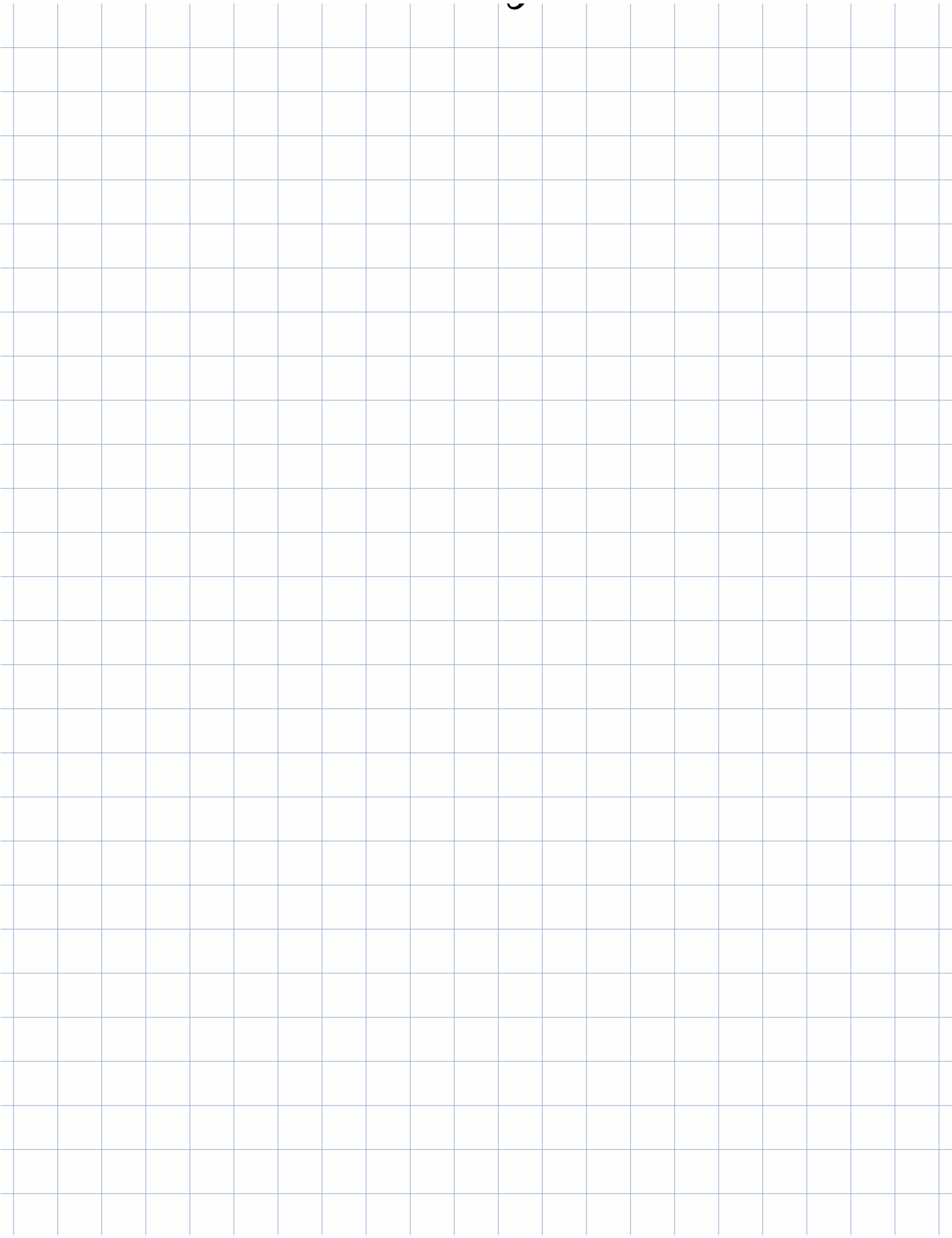
← this is only true if we renormalize

$$\frac{a \langle x, y \rangle + b \langle x, z \rangle}{a+b} \quad ?$$

$$\langle x, y \rangle = \frac{x \cdot y}{|x| + |y|} \rightarrow \frac{x \cdot y}{\sqrt{\|x+y\|}} \quad ?$$

$$\cos \theta = \frac{x \cdot y}{|x| |y|}$$

It'll be something of that form, but I've fiddled enough past answering the exercise



5